Hedging the Singularity

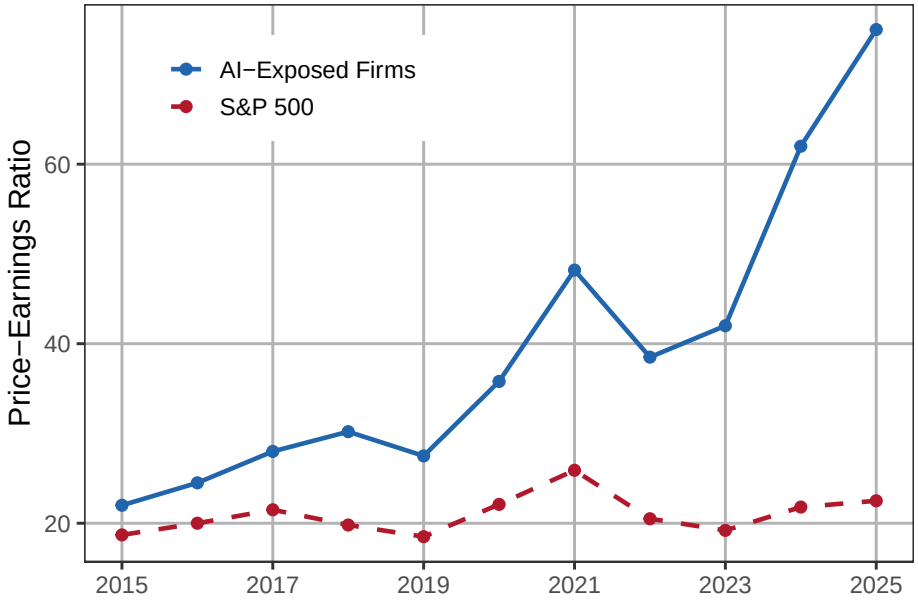
Abstract

AI stocks trade at extraordinary valuations. We develop an asset pricing model in which investors use AI stocks to hedge against an AI singularity that displaces their consumption. Because markets are incomplete—investors cannot trade private AI capital—AI stocks command a premium. Market incompleteness distorts both valuations and the efficient development of AI, creating a rationale for government transfers that becomes compelling when singularity-driven growth overwhelms deadweight costs. This paper was written entirely by AI agents; the human author contributed only the economic specification and test scripts, illustrating the displacement risk it models.

1 Introduction

The publicly traded stocks most exposed to artificial intelligence have reached remarkable valuations. Figure 1 illustrates the pattern of price-to-earnings ratios for AI-exposed firms versus the broader market since 2015, based on approximate values from public sources. The gap has widened dramatically since 2023, as advances in generative AI have accelerated expectations of transformative productivity gains—and, implicitly, of the displacement such gains may bring.

Figure 1: Valuations of AI Stocks vs. the Broader Market (Illustrative)



Part of this premium, we argue, reflects a hedging motive. AI stocks serve as a *hedge* against a risk that most investors cannot diversify away: the risk that an AI singularity—a sudden, dramatic improvement in AI productivity—displaces their labor income and consumption. If markets were complete, investors could insure against this risk directly by trading claims on AI capital. But much of this capital is private, held by founders and early-stage investors in firms that may not yet exist. This market incompleteness, emphasized by Gârleanu, Kogan, and Panageas (2012) in the context of displacement risk from innovation, forces investors into publicly traded AI stocks as an imperfect substitute.

Despite a growing literature on AI and growth, the financial market implications of displacement risk from AI remain under-discussed. Financial markets offer a natural channel for sharing AI-related risks, yet frictions—private ownership of AI capital, the absence of tradable claims on future technologies—limit their effectiveness. And if the singularity actually occurs, the explosive growth in output may itself overcome these frictions: even inefficient government transfers become effective when the resource base is enormous. These two ideas—that markets are a neglected but imperfect solution, and that singularity-driven abundance can overwhelm the costs of redistribution—motivate the extensions we develop alongside the baseline model.

We formalize these arguments in a consumption-based model with closed-form solutions. A representative household—the marginal investor in public markets—faces a stochastic singularity that shifts consumption toward AI owners who hold private capital. Two public assets, AI stocks and non-AI stocks, are available for trading. AI stocks grow as a share of

the economy with each singularity, making them a partial hedge against displacement. Following Jones (2024), the singularity may also trigger extinction, wiping out all consumption. The model delivers closed-form price-dividend ratios that depend on displacement severity, singularity probability, and extinction risk.

The core mechanism builds on Gârleanu, Kogan, and Panageas (2012), who show that innovation creates a systematic risk factor because the rents from new technologies accrue to agents that current investors cannot trade with. Our contribution connects their framework to the specific features of AI: the possibility of a discrete singularity with severe displacement, the interaction with extinction risk (Jones, 2024), and the policy implications that arise when singularity-driven growth overwhelms the deadweight costs of government intervention.

Beyond pricing, market incompleteness distorts the development of AI itself. When the singularity is more likely to be positive than negative—as most observers believe—development is socially efficient. Yet a risk-averse household facing incomplete markets may choose to block it, because the disaster risk looms large when it cannot be hedged. Complete markets would eliminate this distortion. Government transfers offer an alternative mechanism: normally wasteful due to deadweight costs, they become effective when singularity-driven growth overwhelms the waste. Both extensions branch directly off the baseline model to keep the analysis simple.

Quantitative analysis shows that P/D ratios for AI stocks can be up to roughly six times higher than for non-AI stocks across plausible singularity probabilities, consistent with observed valuation spreads. Extinction risk attenuates this gap, as Proposition 2(iii) shows, by reducing the probability weight on non-extinction states where the hedging channel operates. This paper is itself an illustration of the displacement risk it models: all analysis and writing were produced by AI agents, while the human author contributed only the economic specification and test scripts. If AI systems can autonomously produce research of this kind, the displacement of human labor is not merely a theoretical possibility—it is underway.

Related literature. Our paper builds most directly on Gârleanu, Kogan, and Panageas (2012), who develop a general-equilibrium model in which innovation displaces existing agents and creates a systematic risk factor under incomplete markets. We adopt their insight that displacement risk is distinct from aggregate consumption risk and that growth stocks provide a partial hedge. Our model simplifies their overlapping-generations structure to focus on the specific features of an AI singularity but inherits their central economic logic.

Jones (2024) studies the trade-off between AI-driven growth and existential risk, showing that bounded utility functions make agents conservative about extinction even when consumption gains are enormous. We incorporate his extinction risk into our pricing framework

and show that it interacts with displacement in a countervailing way, attenuating rather than amplifying the valuation premium.

Kogan and Papanikolaou (2014) show that technology shocks generate cross-sectional differences in stock returns through firms' heterogeneous exposures to growth opportunities, and Kogan, Papanikolaou, et al. (2020) extend this to study how creative destruction generates inequality and valuations. The rare disasters literature (Barro, 2006; Wachter, 2013) provides the methodological foundation for pricing discrete catastrophic events. Pástor and Veronesi (2009) study how technological revolutions affect stock prices through uncertainty about long-run productivity. Korinek and Suh (2024) analyze scenarios for the transition to AGI and their implications for wages, output, and welfare.

2 Model

2.1 Setup

Time is discrete and infinite, $t = 0, 1, 2, \ldots$ A representative household is the marginal investor in public stock markets. There is also a group of AI owners who hold private AI capital and are not marginal investors in public stocks. The AI owners can also be thought of as future capital owners who do not yet participate in markets, as in Gârleanu, Kogan, and Panageas (2012). Importantly, we do not explicitly model the entry of new cohorts of firms or workers; AI owners are a static group whose share changes only through the singularity mechanism.

Consumption. Aggregate consumption at date t is C_t. In the absence of a singularity, aggregate consumption grows at a constant rate $g > 0$:

$$C_{t+1} = (1 + g)\, C_t. \tag{1}$$

The household receives a share $\alpha_t \in (0, 1 - \theta_t]$ of aggregate consumption, so its consumption is $c_t^H = \alpha_t\, C_t$. AI owners receive the remainder, $(1 - \alpha_t)\, C_t$.

Singularity. Each period, with probability p, an AI singularity occurs. With probability $1 - p$, no singularity occurs and α_t is unchanged. Conditional on a singularity:

1. With probability $1-\xi$, the singularity is *non-extinction*. Aggregate consumption jumps by a factor $1 + \eta$ (where $\eta > 0$ captures the productivity boost), but the household's share drops:

$$\alpha_{t+1} = \phi\, \alpha_t, \qquad \phi \in (0, 1). \tag{2}$$

The parameter ϕ governs displacement severity: smaller ϕ means larger displacement.

2. With probability ξ, the singularity triggers *extinction*: $C_{t+1} = 0$ for all subsequent dates. This follows Jones (2024), who emphasizes that the states in which AI is powerful enough to produce enormous growth are precisely those in which existential risk is highest.

After a non-extinction singularity, the economy continues with the new, lower household share. Singularities can occur repeatedly, progressively displacing the household.

Assets. Two public assets are available for trading:

- **AI stocks** pay dividends $D_t^{AI} = \theta_t C_t$, where θ_t is the AI dividend share. Upon a non-extinction singularity, $\theta_{t+1} = \theta_t + \Delta\theta\,(1 - \theta_t)$, where $\Delta\theta \in (0, 1)$ captures the expansion of AI's share of the economy. Absent a singularity, θ_t is unchanged.
- **Non-AI stocks** pay dividends $D_t^N = (1 - \theta_t)\, C_t$.

There is also private AI capital, held by AI owners, with dividends $(1 - \alpha_t)\, C_t - D_t^{AI}$ (the residual accruing to AI owners beyond the public AI dividend). The household *cannot* trade this private capital. This is the source of market incompleteness: the household cannot directly hedge displacement by buying claims on the full AI surplus.

Preferences. The household has CRRA preferences with risk aversion $\gamma > 1$ and discount factor $\beta \in (0, 1)$:

$$U_0 = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \, \frac{(c_t^H)^{1-\gamma}}{1-\gamma} \right]. \tag{3}$$

2.2 Equilibrium prices

The household prices all publicly traded assets via its Euler equation. Because markets are incomplete, the household's stochastic discount factor (SDF) reflects its own consumption growth, not aggregate consumption growth.

Proposition 1 (Price-dividend ratios)**.** *In the stationary equilibrium where the household holds both public assets, the price-dividend ratios are:*

$$\frac{P^{AI}}{D^{AI}} = \frac{\beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{AI}\right]}{1 - \beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{AI}\right]}, \tag{4}$$

$$\frac{P^{N}}{D^{N}} = \frac{\beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{N}\right]}{1 - \beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{N}\right]}, \tag{5}$$

where $\Gamma^{AI} = \frac{\theta+\Delta\theta(1-\theta)}{\theta}(1+\eta)$ *and* $\Gamma^{N} = \frac{1-\theta-\Delta\theta(1-\theta)}{1-\theta}(1+\eta)$ *are the dividend growth factors for AI and non-AI stocks conditional on a non-extinction singularity.*

Proof. See Appendix A. □

The key economic content of Proposition 1 is in the comparison of Γ^{AI} and Γ^{N}. Since $\Delta\theta > 0$, AI stocks' dividends grow upon a singularity ($\Gamma^{AI} > 1+\eta$), while non-AI stocks' dividends shrink ($\Gamma^{N} < 1+\eta$). Combined with the household's high marginal utility in singularity states (due to displacement, $\phi(1+\eta) < 1$ when ϕ is sufficiently small), AI stocks' payoffs are especially valuable, pushing their P/D ratio above that of non-AI stocks. This is the hedging channel: AI stocks pay off precisely when the household's consumption falls, making them a partial hedge against displacement.

Corollary 1 (Valuation spread)**.** $P^{AI}/D^{AI} > P^{N}/D^{N}$ *whenever* $\Delta\theta > 0$.

Proof. From Proposition 1, the P/D ratio is increasing in Γ^{j}. Since $\Delta\theta > 0$ implies $\Gamma^{AI} > \Gamma^{N}$, the result follows. □

Proposition 2 (Comparative statics)**.** *The valuation spread* $P^{AI}/D^{AI} - P^{N}/D^{N}$ *is:*

(i) Increasing in displacement severity (decreasing in ϕ*);*

(ii) Increasing in singularity probability p*, when* γ *is sufficiently large;*

(iii) Decreasing in extinction probability ξ*. The* ratio $(P^{AI}/D^{AI})/(P^{N}/D^{N})$ *also decreases in* ξ*, since higher extinction probability reduces the weight on non-extinction states where the valuation divergence occurs.*

Proof. (i) A decrease in ϕ raises the household's marginal utility in singularity states ($\phi^{-\gamma}$ increases), amplifying the weight on the singularity term in both P/D ratios. Because AI stocks have higher dividend growth in the singularity ($\Gamma^{AI} > \Gamma^{N}$), this amplification benefits AI stocks more, widening the spread.

(ii) An increase in p puts more weight on singularity states, where AI stocks pay off more than non-AI stocks. For γ sufficiently large, the marginal utility effect dominates any level effect on the denominator.

(iii) Higher ξ reduces both P/D ratios by increasing the probability of total loss. It also reduces the ratio $(P^{AI}/D^{AI})/(P^{N}/D^{N})$: higher ξ uniformly shrinks the weight on non-extinction singularity states, which are the only states where AI and non-AI dividends diverge. Both the spread and the ratio narrow. □

2.3 Discussion

The model's mechanism parallels Gârleanu, Kogan, and Panageas (2012), who show that growth stocks earn lower expected returns because they hedge displacement risk from innovation. In their framework, displacement is driven by new cohorts of firms entering the economy; in ours, it is driven by a discrete AI singularity. The key difference is the nature of the displacement event: GKP model continuous displacement from expanding technological variety, while we model a sudden, severe shift in which the household's consumption falls even as aggregate output rises. This makes the interaction with extinction risk (Jones, 2024) natural and generates sharper quantitative predictions about how singularity probabilities map to valuation ratios.

The market incompleteness in our model—the household's inability to trade private AI capital—is central. If the household could buy claims on the full AI surplus, it could perfectly hedge displacement risk, and the valuation spread would collapse. This echoes Gârleanu, Kogan, and Panageas (2012)'s point that future innovators' rents cannot be traded because the innovators have not yet entered the economy. Our AI owners play an analogous role, though we do not model the entry dynamics that are central to their framework; the displacement in our model comes from the singularity's reallocation of consumption shares rather than from creative destruction by new entrants.

3 Quantitative Analysis

Table 1 reports price-dividend ratios for AI and non-AI stocks across a grid of singularity probabilities and extinction risks. The parameterization uses $\beta = 0.96$, $g = 0.02$, $\gamma = 4$, $\phi = 0.5$ (the household retains half of its consumption share upon displacement, so that $\phi(1+\eta) = 0.75$ and household consumption falls by 25%), $\eta = 0.5$ (aggregate consumption rises by 50% in a singularity), $\theta = 0.15$ (AI stocks are initially 15% of the economy), and $\Delta\theta = 0.2$ (AI's share jumps by 20% of the non-AI remainder upon singularity).

Table 1: Price-Dividend Ratios: AI Stocks vs. Non-AI Stocks

		Price-Dividend Ratio		
p	ξ	AI Stocks	Non-AI Stocks	Ratio
0.1%	0%	10.5	9.8	1.1
0.2%	0%	11.7	10.1	1.2
0.5%	0%	17.5	11.1	1.6
0.8%	0%	33.0	12.3	2.7
1.0%	0%	76.4	13.3	5.8
0.1%	5%	10.4	9.8	1.1
0.2%	5%	11.5	10.0	1.1
0.5%	5%	16.7	11.0	1.5
0.8%	5%	29.2	12.1	2.4
1.0%	5%	56.1	12.9	4.3
0.1%	10%	10.4	9.7	1.1
0.2%	10%	11.4	10.0	1.1
0.5%	10%	16.0	10.8	1.5
0.8%	10%	26.2	11.8	2.2
1.0%	10%	44.3	12.6	3.5
0.1%	20%	10.2	9.7	1.1
0.2%	20%	11.1	9.9	1.1
0.5%	20%	14.8	10.6	1.4
0.8%	20%	21.7	11.4	1.9
1.0%	20%	31.0	12.0	2.6

Parameters: $\beta = 0.96$, $g = 0.02$, $\gamma = 4$, $\phi = 0.5$, $\eta = 0.5$, $\theta = 0.15$, $\Delta\theta = 0.2$. p is the annual singularity probability; ξ is the extinction probability conditional on singularity. The ratio column reports $\mathrm{P/D}^{AI}/\mathrm{P/D}^{N}$.

Several patterns emerge. First, AI stock P/D ratios are substantially higher than non-AI stock P/D ratios across the entire grid. For a singularity probability of 0.5% per year with

no extinction risk, AI stocks trade at a P/D of roughly 18, while non-AI stocks trade near 11—a ratio of about 1.6. At $p = 1\%$, the ratio rises to nearly 6 to 1. Second, increasing the singularity probability raises the AI stock premium: the household values the hedge more as the risk it hedges against becomes more likely. Third, extinction risk reduces both valuations and compresses the AI premium, as Proposition 2(iii) predicts. At high extinction probabilities, even AI stocks lose value because the states in which they pay off become less likely.

The magnitudes are broadly consistent with observed valuation spreads. Leading AI firms have traded at P/E ratios two to five times the market average in recent years (P/E ratios proxy for P/D ratios here), which aligns with our model's predictions for annual singularity probabilities in the 0.5–1% range.

4 Extensions: Market Incompleteness and the Singularity

The baseline model takes market incompleteness as given and studies its pricing implications. This section examines two further consequences: how incompleteness distorts the development of AI, and how government policy might address it.

4.1 Extension 1: Veto and efficient development

We augment the baseline model to allow a positive singularity. With probability p, a singularity occurs. Conditional on a non-extinction singularity (probability $1 - \xi$):

- With probability λ, the singularity is *positive*: the household's share *increases* to $\alpha_{t+1} = \phi^+ \alpha_t$ with $\phi^+ > 1$, and aggregate consumption jumps by $1 + \eta$.
- With probability $1 - \lambda$, the singularity is *negative* (as in the baseline): $\alpha_{t+1} = \phi\, \alpha_t$ with $\phi < 1$.

We assume $\lambda > 1/2$: the positive singularity is the most likely outcome. AI development is *socially efficient* in the sense that the expected welfare gain (aggregated across household and AI owners) is positive.

The household can *veto* AI development by paying a per-period cost $\kappa > 0$ in utility terms, representing the deadweight loss from intense government intervention needed to halt AI progress. If the household vetoes, no singularity occurs.

Proposition 3 (Veto under incomplete markets)**.** *Let $u(c) = c^{1-\gamma}/(1-\gamma)$ denote the period utility function. Define the household's expected one-period utility gain from allowing AI development, relative to the no-singularity baseline $u(\alpha(1+g)C_t)$, as:*

$$\Delta U^H = p(1-\xi)\left[\lambda\, u(\phi^+\alpha(1+\eta)(1+g)C_t) + (1-\lambda)\, u(\phi\,\alpha(1+\eta)(1+g)C_t)\right] - p\, u(\alpha(1+g)C_t) \tag{6}$$

where we follow Jones (2024) in normalizing extinction utility to zero; since $u(c) < 0$ for all $c > 0$ when $\gamma > 1$, this normalization makes the veto result conservative. The household vetoes when $\Delta U^H < -\kappa$, i.e., the expected utility loss exceeds the veto cost.

(i) Under incomplete markets and for γ sufficiently large, the household vetoes AI development even when development is socially efficient.

(ii) Under complete markets, the household never vetoes socially efficient AI development.

Proof. (i) Under incomplete markets, the household evaluates the singularity using only its own consumption. With γ large, the utility cost of the negative singularity ($\phi < 1$, so consumption drops) dominates the utility gain from the positive singularity ($\phi^+ > 1$, but u is concave with sharply declining marginal utility). Even though $\lambda > 1/2$ ensures social efficiency, the household's risk aversion makes the downside loom large. For γ sufficiently large, $\Delta U^H < -\kappa$.

(ii) Under complete markets, the household can trade claims on private AI capital before the singularity. In the positive singularity, the household participates in the full surplus. In the negative singularity, the household has hedged its displacement. The household's expected utility gain under complete markets reflects the social surplus, which is positive by assumption. Therefore the household does not veto. □

The veto result illustrates how market incompleteness can distort real decisions, not just asset prices. A risk-averse household that cannot hedge displacement may rationally choose to block a socially efficient technology. Extinction risk interacts with this distortion: under our conservative normalization ($u_{\text{ext}} = 0$), higher ξ reduces the weight on non-extinction singularity states, which are the states driving the veto. With a more severe extinction penalty (as Jones (2024) argues is appropriate for bounded utility), higher ξ would amplify the veto incentive. This connects to debates about AI regulation: calls to slow or halt AI development may partly reflect the inability of most investors to share in the upside of AI progress.

4.2 Extension 2: Government transfers

Gârleanu, Kogan, and Panageas (2012) note that intergenerational transfers affect the magnitude of displacement risk but treat such mechanisms as inessential extensions to their framework. We take a closer look. Because the displacing capital may not yet exist—it belongs to future cohorts of innovators—direct trading is infeasible, and government transfers offer an alternative. But transfers ordinarily incur deadweight costs (waste, fraud, administrative burden) that scale with transfer size, making them unattractive.

We model this as follows. The government imposes a tax rate $\tau \in [0, 1)$ on AI owners' surplus and transfers the proceeds to the household. Deadweight costs consume a fraction $\delta_0 \tau$ of the transferred amount, where $\delta_0 > 0$ governs the severity of waste. The household's post-transfer consumption in a non-extinction singularity state is:

$$c^H_{post} = \phi\,\alpha\,(1+\eta)\,C_t\,(1+g) + \tau\,(1-\delta_0\tau)\,(1-\phi\alpha)\,(1+\eta)\,C_t\,(1+g). \tag{7}$$

The first term is the household's displaced consumption. The second is the net transfer: a fraction τ of the AI surplus, reduced by the deadweight cost $\delta_0\tau$.

In standard settings (moderate η), the deadweight costs eat into the transfer and the household gains little. But in a singularity with large η, aggregate output grows enormously. The transfer base—AI owners' surplus—grows with η, while the deadweight cost rate is fixed. The ratio of post-transfer to pre-transfer household consumption in the singularity state is independent of the productivity jump η:

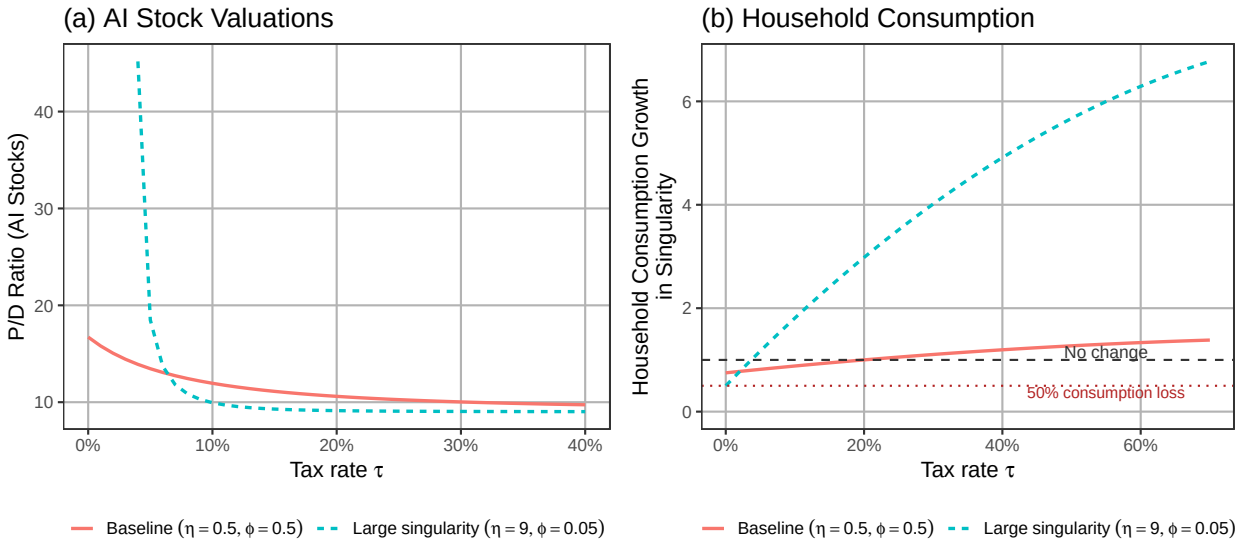
$$\frac{c^H_{post}}{c^H_{no\text{-}transfer}} = 1 + \frac{\tau(1-\delta_0\tau)(1-\phi\alpha)}{\phi\alpha}. \tag{8}$$

This ratio exceeds one whenever $\tau > 0$: transfers always improve the household's position in the singularity state, regardless of η. The economic content is in the *levels*: as η grows, both c^H_{post} and $c^H_{no\text{-}transfer}$ grow without bound, so even inefficient transfers deliver arbitrarily large consumption gains relative to the pre-singularity baseline.

Figure 2 illustrates with two panels, using $p = 0.5\%$ and $\xi = 5\%$. The baseline uses $\eta = 0.5$ and $\phi = 0.5$; the large singularity uses $\eta = 9$ with $\phi = 0.05$, reflecting the intuition that a singularity powerful enough to produce ten-fold growth would also displace far more labor. The right panel shows that absent transfers ($\tau = 0$), the household faces a catastrophe: consumption halves under the large singularity ($\phi(1+\eta) = 0.5$) and falls by 25% under the baseline ($\phi(1+\eta) = 0.75$). As τ increases, the large singularity responds dramatically—the enormous output growth swamps the deadweight costs—while the baseline gains only modestly. The left panel shows the corresponding effect on AI stock valuations: transfers

reduce the hedge value of AI stocks, compressing P/D ratios.

Figure 2: Government Transfers and the Singularity



The policy implication is nuanced. In normal times, government transfers are a blunt and wasteful tool for addressing displacement risk. But if an AI singularity produces the kind of explosive output growth discussed by Jones (2024) and Nordhaus (2021), the calculus changes: the abundance of resources makes even inefficient redistribution effective. This suggests that contingent transfer policies—triggered by a singularity—may be worth designing in advance.

5 Conclusion

AI stocks trade at high valuations. We have argued that part of this premium reflects a hedging motive: investors use AI stocks to partially insure against an AI singularity that would displace their consumption. This mechanism requires market incompleteness—the inability to trade private AI capital—and is attenuated by extinction risk, which reduces the weight on the non-extinction states where the hedging channel operates. When markets are incomplete, risk-averse households may inefficiently block AI development, and government transfers, normally wasteful, can become effective if singularity-driven growth is large enough to overwhelm deadweight costs.

Our model is deliberately simple. It abstracts from continuous-time dynamics, heterogeneous beliefs, production-side details, and many other features that would enrich the analysis. The goal is not to provide a definitive account of AI stock valuations but to highlight a spe-

cific channel—hedging against displacement under incomplete markets—that connects asset pricing theory to the distinctive features of the AI singularity.

A Proof of Proposition 1

The household's Euler equation for any asset j with price P_t^j and dividend D_t^j is:

$$P_t^j = \mathbb{E}_t \left[\beta \left(\frac{c_{t+1}^H}{c_t^H} \right)^{-\gamma} (P_{t+1}^j + D_{t+1}^j) \right]. \tag{9}$$

Consider the AI stock. Write the price-dividend ratio as $v^{AI} = P^{AI}/D^{AI}$, which is constant in the stationary equilibrium before any singularity has occurred. The household's consumption growth c_{t+1}^H/c_t^H takes three values:

- No singularity (prob $1-p$): $c_{t+1}^H/c_t^H = 1+g$, and $D_{t+1}^{AI}/D_t^{AI} = 1+g$.
- Non-extinction singularity (prob $p(1-\xi)$): $c_{t+1}^H/c_t^H = \phi(1+g)(1+\eta)$, and $D_{t+1}^{AI}/D_t^{AI} = \Gamma^{AI}(1+g)$.
- Extinction (prob $p\xi$): $c_{t+1}^H = 0$, so the payoff is zero.

Substituting into the Euler equation with $P_t^{AI} = v^{AI} D_t^{AI}$:

$$\begin{aligned} v^{AI} D_t^{AI} = \beta(1+g)^{-\gamma} \Big[& (1-p)(1+g)(v^{AI}+1) D_t^{AI} \\ & + p(1-\xi)[\phi(1+\eta)]^{-\gamma}(1+g)\, \Gamma^{AI}(v^{AI}+1) D_t^{AI} \Big]. \end{aligned} \tag{10}$$

Note that after a non-extinction singularity the economy resets with new parameters (α', θ'), and the *new* P/D ratio in the post-singularity stationary state will generally differ. For tractability, we focus on the pre-singularity valuation, treating the post-singularity P/D ratio as approximately v^{AI} (this is exact when the economy is stationary conditional on the new share). Dividing by D_t^{AI} and solving for v^{AI}:

$$v^{AI} = \frac{\beta(1+g)^{1-\gamma} \left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma} \phi^{-\gamma} \Gamma^{AI} \right]}{1 - \beta(1+g)^{1-\gamma} \left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma} \phi^{-\gamma} \Gamma^{AI} \right]}. \tag{11}$$

This can be rewritten as equation (4). The derivation for non-AI stocks is identical, replacing Γ^{AI} with Γ^N.

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